

Stream#

<https://pytorch.org/docs/stable/generated/torch.cuda.Stream.html#torch.cuda.Stream>

Description:

Wrapper around a CUDA stream. A CUDA stream is a linear sequence of execution that belongs to a specific device, independent from other streams. It supports `with` statement as a context manager to ensure the operators within the `with` block are running on the corresponding stream. See CUDA semantics for details. `device` (`torch.device` or `int`, optional) – a device on which to allocate the stream. If `device` is `None` (default) or a negative integer, this will use the current device. `priority` (`int`, optional) – priority of the stream, which can be positive, 0, or negative. A lower number indicates a higher priority. By default, the priority is set to 0. If the value falls outside of the allowed priority range, it will automatically be mapped to the nearest valid priority (lowest for large positive numbers or highest for large negative numbers). Check if all the work submitted has been completed.

Function Signature

```
class torch.cuda.Stream(device=None, priority=0, **kwargs)
[source]#
```

Parameters

```
query()[source]#
```

Returns

Return type

Returns:

Recorded event.

Notes & Warnings

NOTE

Note This is a wrapper around `cudaStreamSynchronize()`: see CUDA Stream documentation for more info.

NOTE

Note This is a wrapper around `cudaStreamWaitEvent()`: see CUDA Stream documentation for more info. This function returns without waiting for event: only future operations are affected.

NOTE

Note This function returns without waiting for currently enqueued kernels in stream: only future operations are affected.

Detailed Documentation

Documentation

Wrapper around a CUDA stream.

A CUDA stream is a linear sequence of execution that belongs to a specific device, independent from other streams. It supports `with` statement as a context manager to ensure the operators within the `with` block are running on the corresponding stream. See CUDA semantics for details.

`device` (`torch.device` or `int`, optional) – a device on which to allocate the stream. If

device is None (default) or a negative integer, this will use the current device.

priority (int, optional) – priority of the stream, which can be positive, 0, or negative. A lower number indicates a higher priority. By default, the priority is set to 0. If the value falls outside of the allowed priority range, it will automatically be mapped to the nearest valid priority (lowest for large positive numbers or highest for large negative numbers).

Check if all the work submitted has been completed.

A boolean indicating if all kernels in this stream are completed.

event (torch.cuda.Event, optional) – event to record. If not given, a new one will be allocated.

Wait for all the kernels in this stream to complete.

This is a wrapper around `cudaStreamSynchronize()`: see CUDA Stream documentation for more info.

Make all future work submitted to the stream wait for an event.

event (torch.cuda.Event) – an event to wait for.

This is a wrapper around `cudaStreamWaitEvent()`: see CUDA Stream documentation for more info.

This function returns without waiting for event: only future operations are affected.

Synchronize with another stream.

All future work submitted to this stream will wait until all kernels submitted to a given stream at the time of call complete.

stream (Stream) – a stream to synchronize.

This function returns without waiting for currently enqueued kernels in stream: only future operations are affected.